Fizika, 10 sinf

1. V_0 tezlik bilan xarakatlanuvchi va radiusi R bo'lgan silliq elastik shayba, silliq gorizontal tekislikda yotgan xuddi shunday shayba bilan to'qnashdi.. Natijada birinchi shayba tezligi 2 marta kamaydi.
1) Ikkinchi shayba markazidan birinchi shayba markazi xarakatlangan to'g'ri chiziqqacha d masofani toping.
2) To'qnashishdan so'ng T vaqtdan keyin markazlar orasidagi masofa S ga teng bo'lsa, T ni toping.

Yechim:

1. Энергия сақланиш қонунидан

$$V_1 = V_0 \sin\alpha = \frac{V_0}{2} \quad (1)$$

(1) дан

$$\sin\alpha = \frac{1}{2}, \quad \alpha = 30^0, \qquad d = R$$

Тўнашишдан кейин шайбалар ўзаро перпендикуляр йўналишларда харакат қилади:

$$(V_1T)^2 + (2R + TV_2)^2 = S^2; \qquad T^2(V_2^2 + V_1^2) + 4RV_2T + 4R^2 - S^2 = 0$$

Демак,

$$T = \frac{-4RV_2 + \sqrt{4S^2(V_2^2+V_1^2) - 16R^2V_1^2}}{2(V_2^2+V_1^2)} \qquad T = \frac{\sqrt{S^2-R^2} - \sqrt{3}R}{V_0}.$$

2. Hajmi V ga teng bo'lgan tsilindr devor bilan ikkita qismga ajratilgan. Atrof-muxit bilan issiqlik bilan almashinuv yo'q. Birinchi qismda miqdori 1,5ν va temperaturasi T_1 bo'lgan vodorod, ikkinchi qismda esa miqdori ν , temperaturasi $5T_1/4$ va bosimi boshqacha bo'lgan azot bilan to'ldirilgan. Devor olib tashlandi.
1) Aralashmaning T_2 temperaturasini toping.
2) Aralashmaning P bosimini toping.

Yechim:

2.

U = const

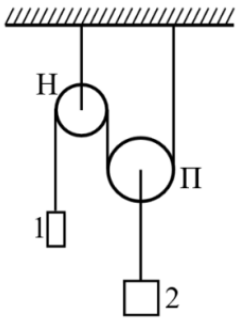
$$1)\ U = \frac{5}{2}R\nu T_1 + \frac{5}{2}R \cdot 1{,}5\nu \cdot \frac{5}{4}T_1 = \frac{115}{16}\nu R T_1$$

$$U = 2{,}5\nu C_{V_{см}} \cdot T_2$$

$$U = 2{,}5\nu \cdot \frac{5}{2}RT_2 = 6{,}25\nu RT_2; \qquad T_2 = 1{,}15T_1$$

$$2)\ P_2 = \frac{2{,}5\nu RT_2}{V} = 2{,}875\frac{\nu RT_1}{V}.$$

3. Tizim rasmda ko'rsatilgandek H, П bloklar va ikkita 1 va 2 –yuklardan iborat.



1-yuk pastga yo'naltirilgan $a_1 = g/5$ tezlanish bilan xarakatlanmoqda.

1) 2-yuk tezlanishini toping.

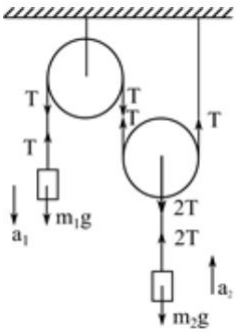
2) 1 va 2- yuklar massalari nisbatini toping.

Iplar, bloklar massalarini, ishqalanish kuchlarni inobatga olmang.

Yechim:

3.

$$a_2 = \frac{1}{2}a_1 = \frac{g}{10} = 1 \text{ м/с}^2 \qquad (1)$$



T – ипга таъсир қиладиган куч бўлса,

$$m_1 g - T = m_1 a_1 \qquad (2)$$

$$2T - m_2 g = m_2 a_2 = \frac{1}{2} m_2 a_1 \qquad (3)$$

(2)-(3) дан

$$(2m_1 - m_2)g = \left(2m_1 + \frac{1}{2}m_2\right)a_1 = \left(2m_1 + \frac{1}{2}m_2\right)\frac{g}{5} \Rightarrow$$

$$\Rightarrow \frac{m_1}{m_2} = \frac{11}{16}$$

@olimpdep

Fan olimpiadalari bo'yicha iqtidorli o'quvchilar bilan ishlash **DEPARTAMENTI**